Plotraa handover: new way of protecting live properties

For: Sant (review, merge, deploy) | From: Pankaj's session with Claude | Date: 19 Oct 2026

Branch: claude/gifted-dirac-onn0ay in BOTH repos (plotra-backend and plotra-frontend)

What changed in one line: a live property no longer shows a blurred picture. Buyers now see the real, unblurred street view, but it is locked to the one spot facing the property: they can look around the house, they cannot walk away from it, and they cannot turn to see the road behind or the back of the house.

1. Old way vs new way

	Before (blurred still)	Now (locked look-around)
What buyers see	One still photo with everything blurred except an oval on the house	Real street view, nothing blurred
Can look around?	No	Yes, within strict limits (section 2)
Can walk down the street?	No	No (arrows, tap-to-move, double-click-to-move all off)
Property coordinates sent to the page	No	No. Only the street-view spot's id and a heading
Before the dealer approves	Movable street view + draggable pin	Unchanged

2. The limits buyers experience (live listings only)

Control	Setting	Where to change it
Turn left / right of the property	50 degrees	LIVE_MAX_TURN, plotra-frontend/src/components/PropertyMapMedia.jsx
Look down	15 degrees	LIVE_PITCH_MIN (same file)
Look up	45 degrees	LIVE_PITCH_MAX
Zoom out	Capped at 1.3 (no wide view of the street)	LIVE_ZOOM_MIN
Zoom in	Up to 3.5	LIVE_ZOOM_MAX
Walking away	Disabled; if anything still moves the view it snaps back to the panorama	linksControl, clickToGo, pano_changed guard

Dragging past a limit snaps the view back. The view starts centred on the house.

3. How it works

- **Backend** (src/services/frontViewService.js, src/controllers/publicListingController.js): for a live listing the public API now returns `streetView: { panoId, heading }`. It comes from Google's free Street View metadata call (radius 60 m, outdoor imagery only), cached 6 hours per listing. The property's own lat/lng and the raw image links are still never sent for live listings.
- **Frontend** (PropertyView.jsx, PropertyMapMedia.jsx): the live page loads Google's panorama by id, points it at the property, and clamps every change of view to the limits above.

- The old blurred front-view image link (/listings/:slug/front-view.jpg) is no longer used by the page. It is left in place and can be removed later.
- If Google has no outdoor imagery within 60 m, `streetView` is null and the street-view section is simply not shown.

4. Who lists, who sees what

- Owners who want to sell do not list themselves: they become seller leads and a verified dealer is assigned. The **dealer** creates the listing on WhatsApp.
- Before approval (pending / awaiting approval): the dealer sees the full movable street view, satellite view and draggable pin, and the page is not indexed by search engines.
- When the dealer approves ("haan"): the listing goes live and the public page switches to the locked street view.
- Staff and dealers still get exact coordinates from the dashboard and listing API (listingController.js). The database keeps the exact pin and the original image links.
- Once live, the pin cannot be dragged (existing server-side block). A correction needs the listing moved back to pending or a manual fix on the backend.

5. Deploy (order matters)

- Merge the **backend** PR first, then on EC2: `git pull && npm ci && pm2 restart all --update-env`. No migration, no new env vars.
- Then merge the **frontend** PR and redeploy it (Vercel). If the frontend goes first, live pages show no street view until the backend is deployed.
- Google Cloud: the browser Maps key is used by the page to load the panorama. Restrict it by HTTP referrer to your site addresses and to the Maps JavaScript API.

6. Test on a real phone after deploy

- Open a **live** listing on iPhone (Safari) and Android (Chrome): the view is sharp, faces the house, and dragging stops at the limits.
- Try to walk: tapping the ground or double-tapping does not move you; there are no road arrows.
- Try to turn around: you cannot see the road behind or the back of the house.
- Pinch to zoom out: it stops at the default; zooming in works.
- Open a **pending/awaiting approval** listing as the dealer: the full movable street view and the draggable satellite pin still work.
- Open a live listing at an address with no street imagery: the page loads normally with no street-view box.
- Tap the expand button: full screen works and the same limits apply.

7. Known limits

- Anyone can still look at that street from other angles on Google Maps itself. This only stops them doing it through Plotraa.
- Without exact coordinates the browser still learns the camera spot's position (a point on the road) from Google. This is the road position, not the property.
- Every live page view now loads Google's map script in the visitor's browser (billed per panorama load). The old blurred image was billed per image fetch on our server, cached.

- Tests: backend 122/122 pass (two new tests for the panorama lookup). The frontend builds; the look-around limits were checked as logic only, not yet on a real device.